A S MOHAMED ARFEEN

AI Engineer | GenAI Specialist | Python Developer

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PROFESSIONAL SUMMARY

AI Engineer specializing in Generative AI and intelligent software systems with hands-on experience building end-to-end AI applications. Proficient in Python, FastAPI, PyTorch, TensorFlow, LLMs, RAG, AI Agents, Computer Vision, and NLP. Experienced in developing scalable, production-ready AI solutions that automate workflows and solve real-world business problems.

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, Java
AI & GenAI: Machine Learning, Deep Learning, LLMs, RAG, AI Agents, Computer Vision, NLP
Frameworks: FastAPI, Flask, PyTorch, TensorFlow, OpenCV, Transformers, LangChain
Web & Databases: HTML, CSS, JavaScript, Tailwind CSS, MySQL, PostgreSQL
Tools & DevOps: Git, GitHub, Docker, Linux, Postman, VS Code
Core Concepts: REST APIs, Backend Development, System Design, Prompt Engineering

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EXPERIENCE

AI Engineer & Co-Founder

A-Zentrix | 2025 – Present

- Co-founded an AI startup focused on developing intelligent software solutions for businesses using Generative AI, Machine Learning, and modern backend technologies.
- Designed and developed end-to-end AI applications using Python, FastAPI, PyTorch, TensorFlow, and Large Language Models (LLMs).
- Built scalable REST APIs, authentication systems, and backend architectures to support AI-powered web applications.
- Collaborated with cross-functional team members to transform business requirements into production-ready AI solutions.
- Led technical architecture, AI integration, and deployment strategy across multiple software products.

PROJECTS

FORCE : Forecasting and **O**perational **R**eadiness through **C**omprehensive **E**mployee-welfare Intelligence

Developed an AI-powered platform to predict personnel stress, burnout, and operational readiness using HRMS, behavioral, and wellness data.

- Built an explainable AI prediction engine using XGBoost, Transformer Models, and SHAP for real-time welfare risk assessment.
- Developed secure role-based dashboards, backend APIs, and a cross-platform wellness application for AI-assisted decision support.
- Architected a scalable, privacy-first platform with RBAC, Zero Trust Security, and enterprise-ready deployment.

Tech Stack: Python | FastAPI | React | Flutter | XGBoost | SHAP | PostgreSQL | Redis | Docker
GitHub: https://github.com/asmohamedarfeen/SIH_v2

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CogniCanvas – AI Coding Agent Workspace

Engineered a visual workspace to monitor and guide AI coding agents across code, terminal, and browser workflows.

- Integrated ACP-compatible agents with configurable model profiles and execution backends.
- Unified Monaco editing, live code diffs, terminal execution, and browser inspection.
- Added conversation branching and MCP tool connections, with web and Electron build targets.

Tech Stack: React | TypeScript | Xterm.js | Zustand | TanStack Query | WebSockets | Docker

GitHub: https://github.com/asmohamedarfeen/CogniCanvas

CortexMesh – Multi-Agent AI & Hybrid RAG Platform

Engineered a multi-agent AI platform that coordinates specialized workflows to retrieve evidence, reason across documents and knowledge graphs, and execute tools with human approval.

- Built a 12-stage LangGraph workflow for intent routing, query rewriting, retrieval, reasoning, and response evaluation.
- Implemented hybrid RAG using ChromaDB, BM25, and Reciprocal Rank Fusion, with NetworkX for relationship-based retrieval.
- Integrated MCP tools, human approval gates, PostgreSQL-backed workflow persistence, and authenticated FastAPI streaming endpoints.

Tech Stack: Python | FastAPI | LangGraph | ChromaDB | NetworkX | PostgreSQL | MCP | Docker | Prometheus | Grafana

GitHub: https://github.com/asmohamedarfeen/CortexMesh

EDUCATION

Bachelor of Technology (B.Tech.) – Artificial Intelligence & Data Science **2024-2028**

Nandha Engineering College (Autonomous), Erode, Tamil Nadu

Affiliated to Anna University

CERTIFICATIONS

Google × Kaggle	— 5-Day AI Agents Intensive Course
LinkedIn Learning	— Advanced Prompt Engineering Techniques
IBM Cognitive Class.	— Machine Learning with Python
SWAYAM (Skill India)	— Machine Learning Using Python Programming
GUVI	— Deep Learning using PyTorch
Great Learning	— Introduction to Neural Networks & Deep Learning

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LEADERSHIP & ENTREPRENEURSHIP

Founder & AI Lead | A-Zentrix

- Founded and led an AI startup, managing a team of 8 engineers to build production-ready AI solutions.
- Drove product strategy, technical architecture, client communication, and end-to-end project delivery.